

Predictive Maintenance Of Heavy Vehicles Using Machine Learning

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Abstract

Predictive maintenance of heavy vehicles is a beneficial method that is widely being implemented by many heavy vehicle companies to greatly reduce operational costs, unexpected vehicle breakdowns or failures and improve overall safety by a great extent. Data collected from the vehicle sensors is further provided to the machine learning techniques to be analysed and predict the failure of the vehicle. Before that, the most important step is to clean, scrub and pre-process the collected data. Pre-processing the sensor data handles missing, inconsistent values, corrupted data and increases the accuracy of the final output. Since, in reality the failure cases will be less compared to normal one, an imbalance in the collected data class is observed. Synthetic Minority Over-Sampling Technique is utilized to balance the class, which helps improve the model accuracy. Several Machine learning algorithms such as, Logistic Regression, Random Forest, Decision Tree, Gradient Boosting, XG boost are trained and tested along with an ensemble Voting Classifier that integrates Random Forest, XGBoost, LightGBM, and Logistic Regression. Model evaluation is conducted using performance metrics such as accuracy, precision, recall, and F1-score along with error-based cost measures. Experimental results show that the Voting Classifier achieves the highest predictive performance with an accuracy of 98.5%, precision of 98.5%, recall of 98.5%, and F1-score of 98.5%, outperforming other models. To make the whole system more understandable to the end-user, explainable AI-models such as LIME and SHAP are implemented to provide visual representations of feature-level explanations. A web application is built using FLASK framework to provide real-time predictions about possible failures.

Keywords

Predictive Maintenance, Machine Learning, Ensemble Learning, XG Boost, Explainable Artificial Intelligence, LIME, SHAP, Flask

1 Introduction

The utilization of heavy vehicles is increasing widely for various use cases such as transportation, Logistics and large-scale goods movement. In continued usage of these vehicles, they are exposed to tough conditions and forced to operate for lengthy hours and carry heavy loads. This prolonged usage exposes the vehicles to mechanical wear and tear which can lead to unexpected failures, sudden need for repairs, and operational delays. Hence uninterrupted monitoring of the vehicle condition is immensely necessary in reducing the cost of vehicle maintenance and improve efficiency and safety [14].

Traditionally, vehicle maintenance is categorized into pro-active maintenance and reactive maintenance. Both the methods have

their drawbacks and prove to be ineffective. While pro-active maintenance repairs the vehicle before a break-down occurs, it fails to take into consideration the current condition of the vehicle and can be unnecessary sometimes. Whereas, reactive maintenance only fixes the vehicle after a breakdown has occurred which can lead to greater down-times [5]. With the help of a trained model and advancement in Artificial Intelligence, it is possible to continuously monitor the vehicle condition. It can also be capable of predicting the vehicle breakdown before it occurs which beats the incapacity of the traditional maintenance techniques [13]. Machine learning models are capable of learning even the slightest changes and patterns in the vehicle performance or engine metrics. This can prove to be beneficial in predicting failures before they occur and greatly reduce unexpected breakdowns and cost to repair them [2]. Predictive maintenance techniques generally take the diagnostic data as input and classify between normal and faulty conditions. To identify and evaluate vehicle health and correctly treat the potential failures, we can make use of supervised classification along with ensemble learning methods and enable preventive measures [8]. Modern approaches also introduce IOT-based monitoring systems that can replicate similar operational conditions to continuously monitor vehicle sensor data [24]. This contributes to overall improvement of the maintenance system and reduce the operational down-time issues [21]. We can also monitor multiple sources of data such as pressure, vibrations and other metrics by integrating advanced algorithms for early detection of potential failures [23].

Additionally, the implementation of user-friendly interface makes it easier for operators to input vehicle sensor data and obtain predictions [1]. Explainable AI usage helps provide visual representation and valid explanation. This aids in having better understanding and transparency regarding the provided prediction and the reason it has been given. It highlights the parameters and metrics that contributed to achieve the obtained prediction [15]. Overall, the aim is to build an efficient system that is capable of continuous monitorship of the vehicle sensor metrics, learn the patterns and provide predictions that help in pre-planning vehicle maintenance, reduce operational costs, reduce down-times and overall improve efficiency [10].

2 RELATED WORK

According to its potential to save on the downtimes, knowledge of improving safety, and efficiency, predictive maintenance has become one of the most notable points of concern in both vehicle and industrial operation today. Innovative approaches to monitoring the state of vehicles and machines have been examined in the recent literature based on the use of data-driven approaches. Omiya, Hanyuda and Nagatomi [17] observed the engine oil degradation

in different vehicles revealing that continuous monitoring of the critical parameters is important to avoid failure and increase the life of the vehicle. Their work also illustrated that combining sensor data with the analysis models can offer helpful information in terms of the maintenance scheduling.

Dinhakaran, Edwin Raja, Velselvi, and Purushotham [4] proposed an IoT-based predictive maintenance framework, which relies on the real-time sensor information in forecasting equipment failures and unplanned breakdowns reduction using intelligent monitoring devices. A hybrid machine learning model of unexpected changes in lithium-ion batteries, which combines Random Forest and feature selection, was proposed by Kumar, Singh, Narayana, Chandrika, Bajaj and Zaitsev [12] to recognize the existence of unfamiliar patterns of degradation and enhance fault detection rate.

The SCANIA component X dataset, the real-life multivariate time-series data on predictive maintenance, introduced by Kharazian, Lindgren, Magnusson, Steinert, and Andersson Reyna [11] allows the relevant models to be efficiently tested under real-life. A machine learning-controlled fault-tolerant motor introduced by Vlachou and Karakatsanis [22] indicates that predictive maintenance can be used in electromechanical. Skoropad et al. [20] used reinforcement learning and predictive analytics to optimize traffic, suggesting that predictive techniques are also used more widely in traffic systems. Gawande [7] combined AI-based predictive maintenance and telematics that make it possible to understand the data in real time and make decisions. The research by Katib, Albassam, Sharaf and Ragab [9] has shown that TinyML and deep learning can be used in real-time anomaly detection in the IoT system observable devices, where lightweight models achieve good results. As demonstrated by Ramkumar et al. [19], predictive analytics can be helpful in the field of predicting CO₂ emissions, and it is relevant in environmental studies. To enhance predictive maintenance in a mining machine, De La Fuente, Radrigan, and Morales [3] established a hierarchical inference model to better predict defects in complicated systems. Zeb and Lodhi [25] stressed the importance of AI that should be used to minimize downtime and enhance efficiency in various industries. All in all, these works have shown the increased application of AI-based predictive maintenance solutions to cars, industrial machines, and IoT applications, which enhance operability and reliability.

3 MATERIALS AND METHODS

The main goal is to build a prediction model that analyses vehicle data and predict potential failures before they occur. The model makes use of the sensor data to evaluate the health of critical components and provide timely solutions. The workflow is initiated with the input of sensor values either manually or through a dataset via a user interface. The input data must undergo a series of data cleaning steps to reduce missing datapoints, inconsistencies and other errors in the collected information. The data is processed; feature engineering is done and is prepared for prediction [16]. Categorical data such as vehicle type and manufacturers are converted into numerical data for better understanding. Various machine learning algorithms are employed to get trained and tested with the provided dataset. These include Random Forest, Decision

Tree, Gradient Boosting, Logistic Regression, and Extreme Gradient Boosting. Additionally, an ensemble Voting Classifier combines Random Forest, XG Boost, Light GBM, and Logistic Regression to enhance predictive accuracy through collective decision-making [18]. Additionally, XAI short for Explainable Artificial intelligence methods are implemented to improve transparency and provide explainability. LIME provides individual explanations for each of the predicted output which helps obtain a better understanding. Whereas, SHAP assesses the contribution of each parameter in obtaining the provided prediction [6]. Finally the whole system is given an interface via a Web application built using Flask framework. This eases the usage of the said model, where users can enter the sensor values and obtain prediction insights. These insights can help support early detection, plan maintenance before-hand and improve the reliability and efficiency of the vehicles.

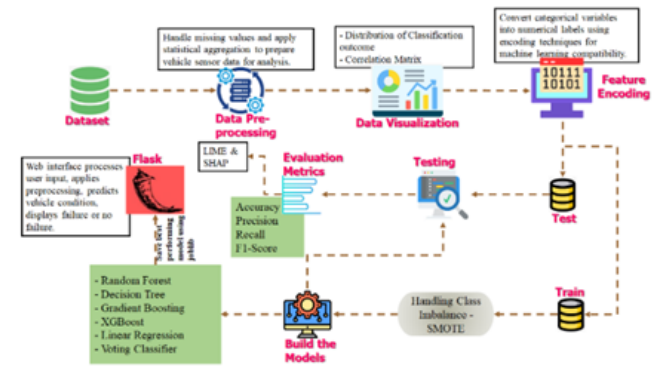


Figure 1: System Architecture

Fig. 1 illustrates a machine learning pipeline for predicting vehicle failures. The process begins with Data Pre-processing and Visualization, followed by Feature Encoding and dataset splitting. After addressing class imbalance with SMOTE, several models are built and tested. Finally, the best-performing model is evaluated using metrics like F1-Score and deployed via a Flask web interface.

3.1 Dataset Collection

The predictive maintenance dataset used as shown in Fig. 2 is a high dimensional time series based on a multivariate structure of multivariate time series is aimed at capturing all the operational behaviors of heavy vehicles. It includes 34 columns which follow particular vehicles and changes through time in sequential time intervals with the combination of both fixed and moving information. Categorical identifiers, including; manufacturer (MAN), engine type, vehicle model are used as part of a static metadata, which give context to individual units. The operational parameters taken by dynamic telemetry are real time measures such as engine power (in kW) and payload capacity (in tons) and anonymized sensor values, such as sensor 12, sensor 13, sensor 14, sensor 15, and sensor 16. There are also also binary target variables in the dataset, failure and event occurred which specify whether the time steps had mechanical faults or anomalies. With metadata, sensor measurements, and labeled results, this collection of data can be used to train supervised learning methods designed to determine

patterns of vehicle behavior that are antecedents to failure events. The architecture will be helpful in training and assessing machine learning models to predictive maintenance, allowing detection of small anomalies in sensor readings that can presuppose imminent failures. Fig. 2 gives out a glimpse of the dataset, which shows how sequential records of vehicles, together with the variety of the features, allows building very strong classification models to the proactive planning of maintenance. This data is the base of data preprocessing, feature engineering and predictive modelling of the maintenance pipeline.

vehicle_id	time_stamp	manufacturer	engine_type	region	vehicle_age	engine_power_kw	payload_capacity_tons	failure	meant_occurred	...	sensor_52	sensor_53	sensor_54	sensor_55	sensor_56	...
1	1/1/2020	0	MAN	Distal_A	Midwest	15	360.05	23.47	0	0	-0.007348	-1.186345	-0.103532	-0.346863	0.284797	-0.9
1	1/2/2020	1	MAN	Distal_A	Midwest	15	360.05	23.47	0	0	-0.240525	-1.127439	-0.460305	-0.988024	1.758246	-1.3
1	1/3/2020	2	MAN	Distal_A	Midwest	15	360.05	23.47	0	0	1.468991	-0.281438	-0.004113	-0.941024	1.017875	-0.4
1	1/4/2020	3	MAN	Distal_A	Midwest	15	360.05	23.47	0	0	1.451788	-0.357868	-1.191038	-0.757183	0.413276	-1.3
1	1/5/2020	4	MAN	Distal_A	Midwest	15	360.05	23.47	0	0	-0.217639	-0.784768	1.488375	-0.073641	1.348875	0.9

5 rows × 16 columns

Figure 2: dataset

3.2 Pre-Processing

Pre-processing of data is the most crucial part in training the model as the data obtained from the vehicle sensors is not usually clean and ready to be used for model training. Before analysing the data, the inconsistencies of the data are identified. Imputation techniques are applied ensuring there are no gaps in the dataset. Following this, feature engineering is applied through statistical aggregation, which summarizes sensor readings and operational metrics into meaningful representations. Statistical aggregation is performed to calculate measures such as mean, median, mode and standard deviation. These measures further help the model in learning the patterns and trends better to improve prediction accuracy.

Data Visualization: Visual representation of the data helps identify underlying trends. The scattering of the data points in the graph helps obtain an insight of the class distribution and identify any class imbalances. The building of a correlation matrix is conducted to get an insight into the relationship between sensor values and car indicators. The matrix gives a pictorial graphic on the readings that were positive or negative in terms of their relationship with each other.

Feature Encoding: It transforms the categorical data into quantitative by feature encoding. The label encoding is used to assign special numbers to the categorical variables like engine type, manufacturer and vehicle model with the same sort of procedure being applied to all the data points. The feature encoding assists the model in extracting the meaning of the qualitative data and preserving the relationships of the original data with not introducing way to anomalies and biases. These encoded attributes are then added to the rest of the information to create a completed dataset is prepared to the model.

Handling Class imbalance: Class imbalance happens when failures are much less than the regular operation records which may skew the machine learning models towards the majority class. In response, the Synthetic Minority Oversampling Technique (SMOTE) is used that creates the synthetic samples of the failure cases of minorities. This normalizes the dataset hence enhancing the capability of the model to identify uncommon failures. Imbalance leads to more positive results during classification, fewer false negative,

and more reliable predictive maintenance system, which may give the opportunity to detect the vehicle breakdown and respond to the situation quickly.

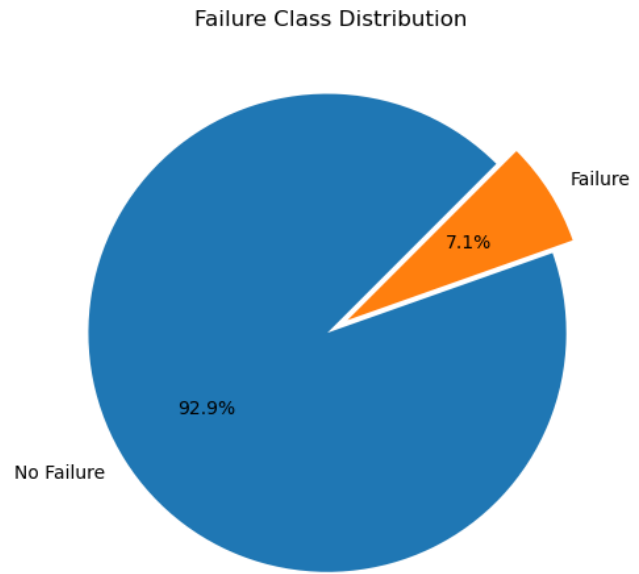


Figure 3: Class Imbalance in Dataset (Failure vs Non-Failure)

Fig.3 shows a strong class imbalance in the dataset, where non-failure instances dominate over failure cases. Such imbalance can negatively affect model performance, especially in detecting rare failure events, hence SMOTE is used to balance the classes.

Train and test split: The dataset is then divided into training and testing splits. The training data is exclusively provided to the machine learning model, so that it learns the patterns and relationships. It helps them understand how various sensor readings affect the failure of the vehicle. The testing data is not revealed to the model so that it can provide unbiased prediction on new datasets. This also prevents overfitting of the model on new data and improve prediction accuracy. In this case, the most common split that is, 80 % training and 20 % testing data is applied. This gives sufficient data in both cases for the model training.

3.3 Algorithms

Random Forest: Random forest is a combination of multiple decision trees with each one having its own prediction. Random forest can be classified as an ensemble learning technique. It works by building multiple decision trees and are combined to produce the final prediction. Each tree takes a random subset of the training data along with selecting a random section of features to predict. This helps in increasing the diversity of the model as well as reducing overfitting of the model on new data. Finally, the trees are combined through majority voting and the overall prediction is produced. In predictive maintenance for heavy vehicles, it analyses the vehicle sensor values, identifies relation between the data sample,

conditions and potential failure cases. This increases the prediction accuracy, reliability and efficiency thoroughly.

Decision Tree: Decision tree is a pretty straight-forward method that constructs hierarchical tree-like structures which represents decisions and their possible outcomes. It is a supervised learning method, in which dataset is split into smaller subsets. Based on the features and operational data, the subset is further built into internal nodes, branches and leaf nodes. The internal nodes consist of the decision rule that is applied to a specific feature whereas, the leaf nodes form the final prediction. The data is continuously split into smaller subsets until the data is sufficiently homogenous or any pre-defined halt conditions are satisfied. Decision trees are far simpler to understand, they provide visual representation of how input features are influencing final outcomes, which helps obtain a clear understanding of the relations. In heavy vehicle predictive maintenance, it analyses the input dataset, learns the patterns and understands the relation between the operational conditions and potential failures.

Gradient Boosting: Gradient Boosting is an ensemble, a model that creates a powerful model by stacking weak learners in series. All the models rectify the mistakes of the earlier models and enhance their performance with time. It reduces the error of prediction by means of gradient descent and works well with the more complex patterns in structured data. It is used in predictive maintenance whereby sensor information is analyzed to determine possible vehicle breakdowns.

Logistic Regression: Logistic Regression is a classification technique that is characterized to forecast the likelihood of an outcome which is categorical. It employs logistic function to map the outputs between 0 and 1 and transforms it into the class labels according to a threshold. It is easy to follow and understand, and thus it can be used in binary classification. In this regard, it predicts the probability of vehicle breakdown using sensor data.

XGBoost: XGBoost is a better variant of gradient boosting that enhances speed and efficiency. It generates decision trees in a series, regularizes to avoid overfitting and is efficient with large datasets. It can learn through past mistakes to detect concealed patterns in sensor data that can predict the failure of the vehicle in an accurate way. **Voting Classifier:** Voting Classifier is the model that fuses the results of several models in order to create a final outcome through majority voting or by averaged probabilities. This solution enhances stability and accuracy through mitigating the weaknesses of individual models. It combines Random Forest, XGBoost, Light-GBM, and Logistic Regression on this system to increase failure prediction.

XAI + Flask: XAI methods like LIME and SHAP are applied to enhance transparency by demonstrating the contribution of each feature to predictions. Individual prediction is expounded using LIME, whereas the overall feature importance is available using SHAP. This is because the system is accessible and easy to use, allowing the users to enter sensor data and get real-time predictions in a Flask-based web interface.

3.4 CRISP-DM Methodology

The proposed project will use the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework to have an organized, effective and reproducible method of predictive maintenance of

heavy vehicles. The methodology allows developing it systematically, at the same time keeping the operational goals and machine learning implementation highly aligned.

1. **Business Concept:** The main aim is the creation of a sound predictive maintenance system that will be able to predict possible heavy vehicle breakdowns beforehand. The formulation of the problem can be represented as a binary classification problem, in which the model will predict the probability of failure by using sensor data. Among the main objectives are the reduction of false negatives (missed failures), minimization of operating costs, and the enhancement of the safety of vehicles.

2. **Data Knowledge:** The project will make use of a predictive maintenance dataset with sensor readings of vehicles. The purpose of the Exploratory Data Analysis (EDA) is to get acquainted with the distribution of features, identify gaps in them, and determine relationships between variables. Data structure and features relationship: Distribution plots and correlation matrices are used as a means of visualizing data.

3. **Data Preparation:** Data preprocessing is very important in enhancing the performance of a model. SimpleImputer is used to deal with missing values to make the dataset complete. The feature engineering is conducted by statistical aggregation which improves the feature representation. Categorical transformation is done through label encoding. In order to solve the imbalance in classes, the SMOTE (Synthetic Minority Over-sampling Technique) technique is applied, which greatly enhances the model learning of the cases of minority failures.

4. **Modeling:** Several machine learning models are applied such as Random Forest, Decision Tree, Gradient Boosting, Logistic Regression, and XG Boost. Also, the ensemble Voting Classifier (RF, XG Boost, Light GBM, Logistic Regression) is built, which takes advantage of the performance of each model. You are hyperparameter tuned and comparatively analyzed, which makes sure that you are choosing the best model.

5. **Evaluation** Model performance is measured by the accuracy, precision, recall, and F1-score as well as cost-based measures (false positives, false negatives, total cost). The Voting Classifier is a low-cost generalizer and reliable classifier, as it is the highest performing (98.5 % on all major metrics). Model effectiveness is also verified with the help of confusion matrices and comparison graphs.

6. **Deployment:** The last model is deployed with the help of Flask-based web application, and it allows real-time prediction. The system accepts user input on the sensors of the vehicle, is used to apply the identical preprocessing pipeline to the input, and presents failure predictions. On top of the SQLite-based user authentication system, the solution is practical and scalable because it ensures access security.

7. **Explainability and Interpretability:** Explainable AI (XAI) methods, including LIME and SHAP, are used to increase transparency. These approaches provide explanations at feature-level, which allows users to learn about the model decision-making, and it boosts confidence on the system.

Iterative Nature of CRISP-DM

CRISP-DM is a process that is by nature iterative. Evaluation and explainability insights are used to make refinements to preprocessing and feature engineering and model selection to make sure that a final solution is continuously improved and robust. The systematic

use of CRISP-DM is that it makes the project methodologically sound, interpretable, and deployment-ready. The system combines sophisticated machine learning with a stepwise process to provide an extremely reliable and useful predictive maintenance system to heavy vehicles.

4 DISCUSSION

The findings of the proposed preventive maintenance system prove that machine learning models are efficient to process information on vehicle sensors with the intention to detect possible failures. The Voting Classifier is the best model when compared to all the other models considered since it was the model that performed the best in all the evaluation metrics, which implies that combination of multiple models will result in increased prediction reliability. SMOTE was used to manage the imbalance of classes, with the models being able to identify failure cases that are rare. Besides, explainable AI methods (LIME and SHAP) are implemented, which improves transparency by giving insight into the impact of various features on predictions. On the whole, the system demonstrates good opportunities to assist in the proactive maintenance choices and to minimize unexpected failures.

5 LIMITATIONS

The system, even though it has high accuracy, has some limitations. The quality and size of the dataset are critical to the model performance, and it might not be effective with the unseen information of a dissimilar type of vehicles and operating environment. Use of synthetic data via SMOTE although useful might not reflect fully on the real-life failure situations. Moreover, the system does not use real time data but uses static data which restricts it to continuous monitoring. Ensemble models could also be computationally complex and consequently raise processing time. The limitations can be resolved through future advancements by adding real-time data provided by the IoT, using larger datasets, and more advanced models.

6 EXPERIMENTAL RESULTS

6.1 Performance Metrics

Accuracy: The accuracy of a test is its ability to differentiate the patient and healthy cases correctly. To estimate the accuracy of a test, we should calculate the proportion of true positive and true negative in all evaluated cases. Mathematically, this can be stated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (1)$$

Precision: Precision evaluates the fraction of correctly classified instances or samples among the ones classified as positives. Thus, the formula to calculate the precision is given by:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (2)$$

Recall: Recall is a metric in machine learning that measures the ability of a model to identify all relevant instances of a particular class. It is the ratio of correctly predicted positive observations to

the total actual positives, providing insights into a model's completeness in capturing instances of a given class.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

F1-Score: F1 score is a machine learning metric that combines both recall and precision scores to measure the model's accuracy. The accuracy metric computes how many times a model made a correct prediction across the entire dataset.

$$\text{F1 Score} = \frac{2 \times (\text{Recall} \times \text{Precision})}{\text{Recall} + \text{Precision}} \times 100 \quad (4)$$

Total Cost: A cost-sensitive metric assigning a penalty weight of 1 to each False Positive and 10 to each False Negative, reflecting that missing a real failure is far more costly than a false alarm. The formula is:

$$\text{Total Cost} = (1 \times \text{FP}) + (10 \times \text{FN}) \quad (5)$$

Where FP = False Positives (weight 1) and FN = False Negatives (weight 10), penalizing missed failures more heavily than false alarms.

6.2 Performance Evaluation Table

Model	Acc	Prec	Rec	F1	FP	FN	Cost
Random Forest	0.966	0.967	0.966	0.966	18	16	178
Decision Tree	0.797	0.918	0.797	0.843	179	24	419
Gradient Boosting	0.973	0.973	0.973	0.973	13	14	153
XGBoost	0.980	0.979	0.980	0.980	7	13	137
Logistic Regression	0.883	0.941	0.883	0.904	103	14	243
Voting Classifier	0.985	0.985	0.985	0.985	8	7	78

Table 1: Performance Evaluation

Table 1 presents the performance evaluation of various machine learning models, comparing accuracy, precision, recall, F1-score, errors, and total cost.

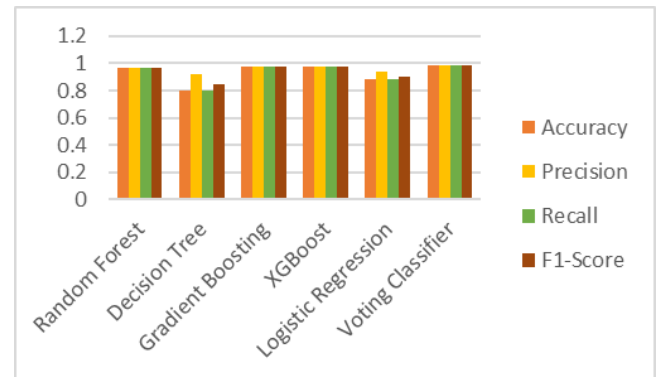


Figure 4: Comparison Graph

The bar chart in Fig. 4 compares evaluation metrics across six machine learning models, highlighting the Voting Classifier's top performance.

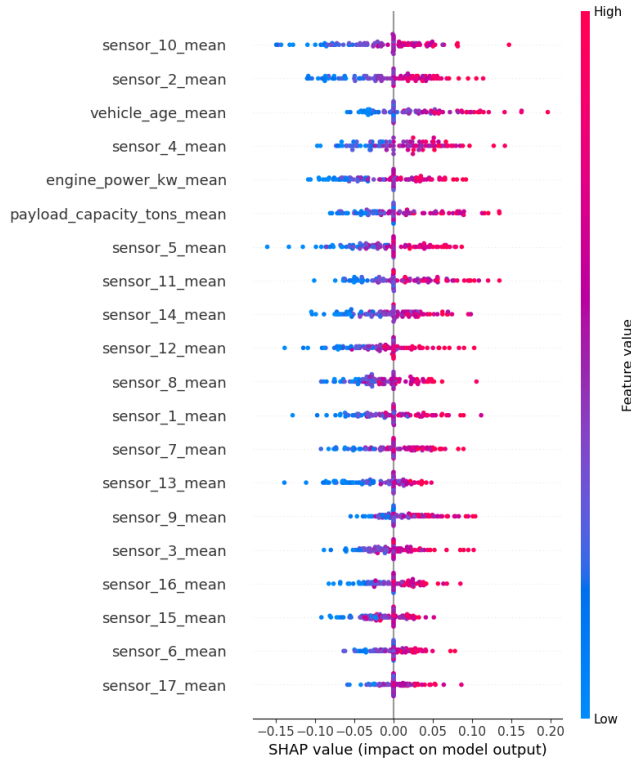


Figure 5: SHAP feature Importance (Voting classifier)
Fig. 5 represents the SHAP beeswarm plot of Voting Classifier.

7 Conclusion

Mechanical failures in heavy vehicles cannot be predicted precisely, so the process of enhancing reliability, decreasing the cost of maintenance, and preventing the emergence of unexpected failures is necessary. This paper introduces a predictive maintenance scheme based on machine learning, which is used to process sensor data on a vehicle to identify failures at early phases. It involves preprocessing of data, feature aggregation, encoding and classification imbalance by use of SMOTE to enhance the performance of the model. Several classification algorithms had been tested, and the Voting Classifier showed the best performance with accuracy, precision, recall, and F1-score of 98.5 percent, which is very effective to check on potential faults and the errors were kept to the minimum level. The Voting Classifier is superior because it can integrate the use of multiple models and this limits the bias and variance of single models thus enhancing the generalization. LIME and SHAP are explainable methods that are combined to help visualize the effect of sensor features on predictions and enhance transparency and trust. The web interface (built in Flask) can be used to make real-time predictions based on sensor values where the user can enter values and get real-time results. On the whole, the system contributes to the effectiveness of the maintenance planning and the improvement of the reliability of the heavy vehicles operations. The future development can comprise the introduction of real-time IoT-based data to monitor the process continuously, application of more sophisticated deep learning algorithms to identify complex patterns, capability

to expand datasets to enhance the generalization, and implement it with cloud or edge computing to make scalable and low-latency predictions. Better decision-making can be also facilitated with the help of enhanced visualization and explainability tools

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